

The Valuation Engine

USER'S MANUAL — THE TERMINAL AT MYOPICDELIRIUM.COM/INSTRUMENTS/TERMINAL

An independent mark on a private holding, with its uncertainty carried honestly all the way to your screen. This manual explains every input, every control, and every number the engine returns — and, just as carefully, what none of it means.

Engine seed 20240115 · every figure in this document was computed by the engine at build time · July 2026

Contents

Contents	2
Read this first	4
The fastest possible orientation	4
1 • The instrument	5
1.1 • What this is	5
1.2 • What this is not	5
1.3 • Getting in: the access code	5
1.4 • The first press, honestly	5
2 • Quick start — the example book, read end to end	6
3 • Describing a holding	7
3.1 • The company	7
3.2 • The last round (optional, but load-bearing)	7
3.3 • The cap table	7
3.4 • Your stake and the levers	9
3.5 • The market reference — the one rule that matters	9
4 • Books, controls, and reproducibility	10
4.1 • Building a book	10
4.2 • Export and import — no accounts, by design	10
4.3 • The three stresses	10
4.4 • Calibration set, seed, and paths	10
5 • Reading the answers	11
5.1 • The band, seriously	11
5.2 • The four headline cards	11
5.3 • The spread and the sensitivity	11
5.4 • The J-curve	12
5.5 • The risk view	13
5.6 • Method notes and the oracle	13
6 • When the engine says no	14
7 • What the engine computes — one honest page	14
8 • Boundaries: public, private, and the gate that is not this gate	15

9 · The owner's page	15
Appendix A · The book file format	15
Appendix B · Glossary	15

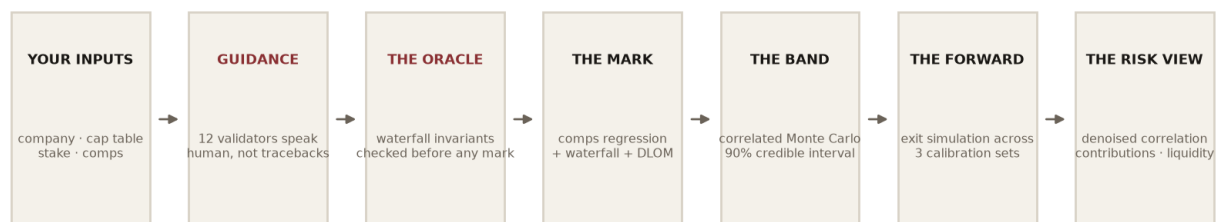
Read this first

The engine never gives you a number. It gives you a **band** — a low, a central, and a high, at a stated credibility — and where it speaks about the future it gives you a **range across disagreeing calibrations**. This is not a display preference. The response formats have no field in which a lone number could travel; a bare mark is unrepresentable. If you remember one thing from this manual, remember why: **a private mark pretending to be a point is the polite name for a lie**, and this instrument was built to not tell it.

Everything is computed on the server. The page you use is a renderer; it holds no model, no shortcut, no secret. Same inputs, same seed — bit-identical answer, every time, forever. That reproducibility is a design contract, not a best effort.

The fastest possible orientation

1 · Open myopicdelirium.com/instruments/terminal and enter the access code. 2 · Press **Load the example book**, then **Run the book**. The answer returns in about a second — three fictional holdings, already computed and cached. 3 · Read the four headline cards left to right: the NAV band, the median TVPI, the IRR target-hit *range*, and the effective bets. 4 · Open a holding's payoff curve from the table. Everything else in this manual is depth behind those four steps.



Everything right of GUIDANCE happens on the server. The browser renders; it never computes.

Figure 1 — the path from your inputs to the answer. Two stations are guards: guidance rejects incoherent inputs in plain language, and the oracle re-proves the waterfall's invariants on your exact cap table before any mark is attempted.

1 • The instrument

1.1 • What this is

The Valuation Engine marks minority stakes in private, venture-backed companies. You describe a company, its capital structure, and your position; you supply a market reference you trust; the engine returns an independent mark with its credible band, the chain of reasoning behind it, and a forward simulation of outcomes. At book scale it aggregates holdings under a correlated Monte Carlo and adds a risk view: who carries the risk, how correlated the book really is, and when it turns liquid.

Two surfaces share the same engine. The **demonstration** at /instruments is a narrative with precomputed output — it always works, needs no code, and computes nothing in your browser. The **terminal** at /instruments/terminal is the live instrument this manual is about.

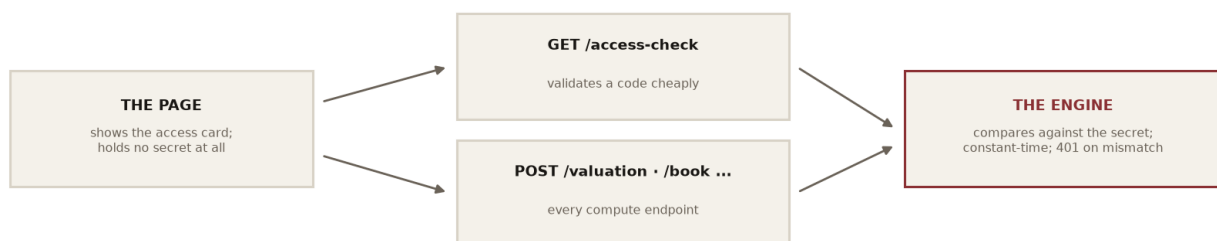
1.2 • What this is not

It is not investment advice, and it is not an oracle. The engine's honesty layer exists precisely because valuation of illiquid assets is *irreducibly* uncertain: the instrument's job is to compute what follows from your inputs and stated assumptions, and to show you how much the answer moves when the assumptions do. It will never hand you false precision to feel decisive with.

The public terminal is also not the venue for marks of record on a real portfolio. That use sits behind a separate, private deployment and a methodology gate (independent validation, IPEV alignment, vendor data QA) that has deliberately not been waived. See section 8.

1.3 • Getting in: the access code

The terminal is featured publicly but computes only with the code. When the page loads it asks the engine whether a gate is on; if so, you see the access card. Enter the code once — it is kept for the browser session only (closing the tab forgets it) and attached to every computation. The check happens on the server with a constant-time comparison: without the code there is nothing a clever visitor can do with the page, because the page can do nothing.



the code travels as the header `x-pve-access-code` on every call; a 401 clears it and re-shows the card

Figure 2 — the access gate. The page holds no secret; the engine refuses codeless compute with a 401. If the code is rotated while you hold the old one, your next press returns you to the card with a message.

If you were not given the code: request it via the site's Connect page. The demonstration needs no code and shows the full instrument on precomputed output.

1.4 • The first press, honestly

The engine sleeps when nobody is using it (that is why it costs almost nothing to keep). The first visit after a quiet spell pays roughly twenty seconds of wake-up; the terminal says so on the button rather than pretending to be slow hardware. The example book is pre-warmed at every deploy, so the recommended first press — **Load the example book**, then run — answers in about a second even cold. Any book you have run before, at the same seed and controls, returns instantly and *identically* from the cache.

2 · Quick start — the example book, read end to end

The example book is three fictional holdings, chosen so their capital structures answer differently: a participating-capped Series B (Norrskan Health), a plain Series A (Fjordpay), and a three-round stack with a 1.5x senior preference (Kompass Logistik). All three run on pasted comparables — the honest mode — so nothing in the example is flagged illustrative. Loading it fills the book shelf and pins the seed; running it produces, verbatim:

holding	mark band (M)	central	DLOM	gap vs self-mark	peer set
Norrskan Health AB	6.90 - 11.86	9.34	44%	+33.4% (comps)	NH1, NH2, NH3, NH4, NH5
Fjordpay AS	2.58 - 6.90	4.45	53%	-19.2% (dlom)	FP1, FP2, FP3, FP4
Kompass Logistik GmbH	8.46 - 11.05	9.62	36%	-19.8% (dlom)	KL1, KL2, KL3, KL4, KL5

Table 1 — the example book's holdings table, as the terminal shows it. The DLOM column is the liquidity discount the engine applied; the gap column compares the engine's central to the self-mark carried in the example.

The book-level cards read: NAV band **20.2 - 27.3M** (central 23.7M, 90% credible interval); median forward TVPI **1.39x** with CI [0.47, 3.29]x; P(IRR at or above 15%) **12% - 41%** across calibration sets. If your screen shows these numbers at seed 20240115, your terminal and this manual were computed by the same engine.

Reproduce-or-distrust is the intended posture: every quoted number in this manual is regenerated from the live engine when the manual is built.

3 · Describing a holding

Units, once: all monetary amounts are in **millions**, and share counts are in millions of shares — so a price per share reads in whole currency units. Growth rates and margins are decimals (0.35 means 35%). Blank means "not known": the engine substitutes nothing silently; where an absence matters it says so in the method notes.

3.1 · The company

field	meaning	what it drives
name	display label	nothing numeric — label only
sector / stage / geography	categorical identity	stage sets the DLOM default and exit-type mix; sector routes the risk-view proxy
revenue (LTM)	last-twelve-months revenue, M	the primary comps multiple (EV/Revenue) applies to this
growth (YoY)	revenue growth, decimal	positions the company inside the peer regression
gross margin	decimal 0-1	quality adjustment in the comps regression
EBITDA (LTM)	optional, M (negative fine)	enables the EV/EBITDA leg of the blended multiple
cash / burn	balance M · monthly burn M	runway context; stressed exits interact with it

Table 2 — company fields.

3.2 · The last round (optional, but load-bearing)

Pre-money, post-money, and amount raised of the most recent priced round. The exit simulation anchors its distribution of exit equity values on the last-round post-money; without it, the engine falls back to the comps-implied equity value and records that fallback in the method notes. Enter any two of the three and the intake layer checks their arithmetic against each other rather than trusting one.

3.3 · The cap table

One row per share class, top of the stack first. **Rank 0 is paid first.** The engine enforces coherence — duplicate ranks, a preference on common, shares of zero, or an owned class that does not exist all come back as plain-language guidance, never a stack trace:

· *Series B: Shares outstanding must be greater than 0.0.*

column	meaning
class / type	name; preferred or common
rank	seniority: 0 is paid first; ties are pari passu and share pro-rata within the tranche
pref x	liquidation preference multiple on invested capital (blank or 0 for common)
part.	participating preferred: takes its preference AND shares in the remainder
cap x	participation cap as a multiple of invested capital; blank = uncapped
shares (M) / invested (M)	shares outstanding and capital invested for the class

Table 3 — cap-table columns.

As you type, the terminal redraws **your payoff curve**: proceeds to the class you hold as a function of the company's equity value at exit. This curve is computed exactly — kinks found by bisection to machine tolerance, conversion decided at a fixed point, participation caps water-filled — not sampled. It is the single most useful diagnostic while entering a structure: if the curve does not look like your term sheet reads, the cap table is wrong.

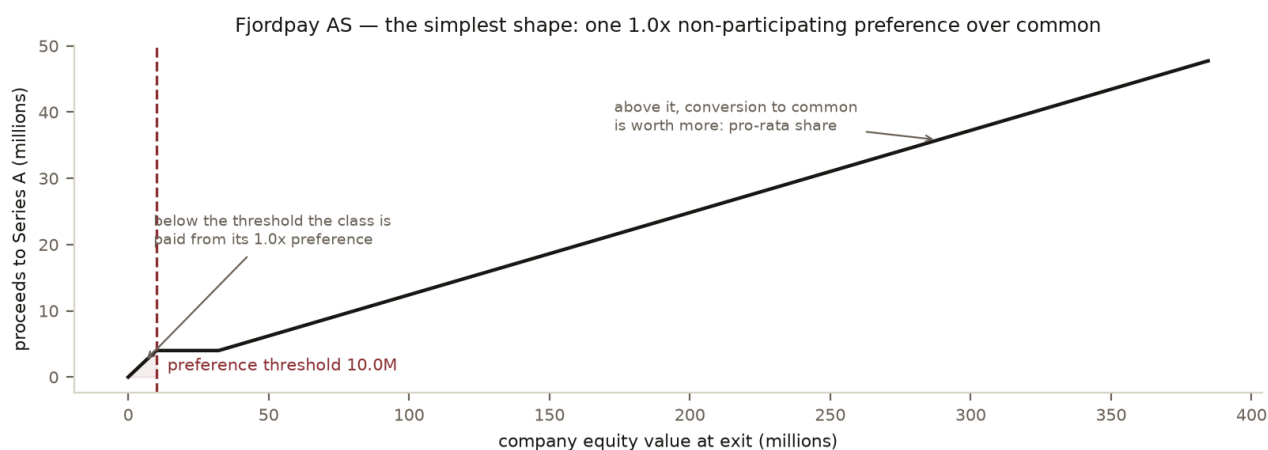


Figure 3 — the simplest payoff shape (Fjordpay). The dashed line is the preference threshold shipped with every curve: the equity value where the stack's preferences are exactly covered.

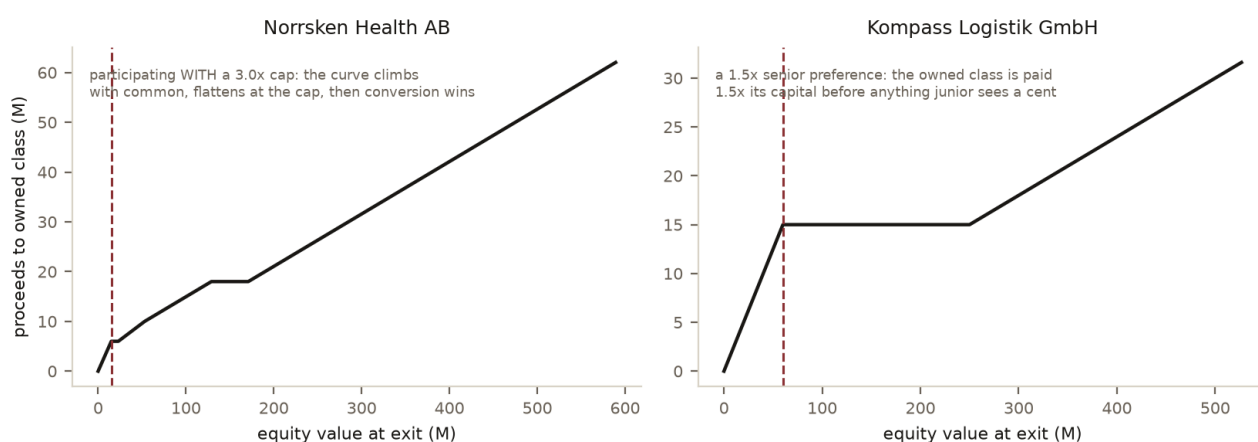


Figure 4 — two structures that punish intuition. Left: participation with a 3.0x cap produces a flat shelf — between the cap and the conversion point, extra exit value goes entirely to others. Right: a 1.5x senior preference moves the whole stack's threshold right; junior classes see nothing until it clears.

3.4 · Your stake and the levers

The stake block names the class you hold, your shares, and your cost basis. Two optional entries change what you get back rather than what is computed: a **self-mark** (what you currently carry the position at) switches on the gap decomposition — the engine tells you how much of the disagreement comes from the market reference and how much from the liquidity discount, and names the dominant driver. Two **levers** override engine defaults: the DLOM central (0 to 0.85) and the revenue-vs-EBITDA blend weight (0 to 1).

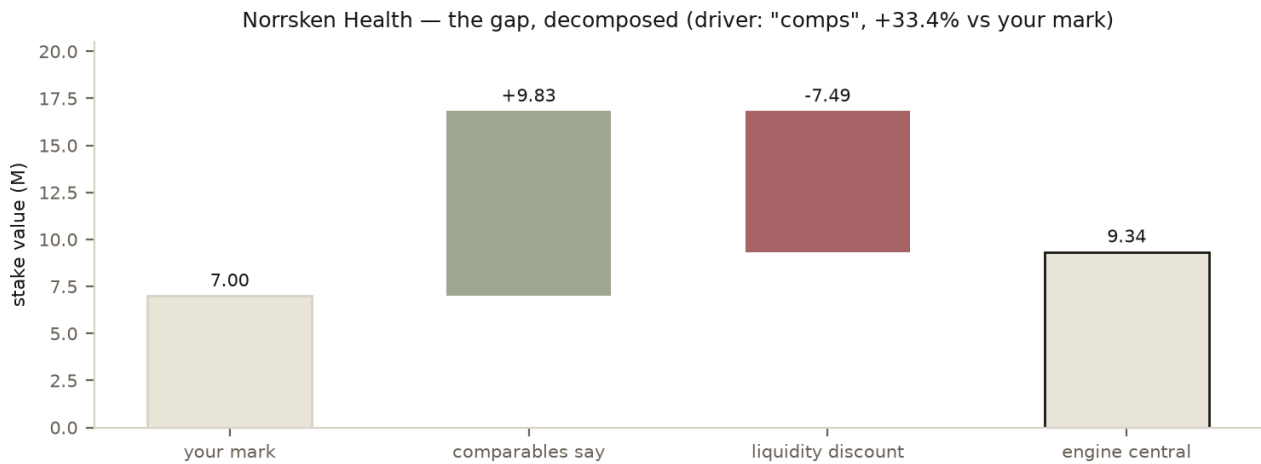


Figure 5 — the gap decomposition for Norrskan Health, from the example book's self-mark to the engine's central.

On the DLOM default: the engine prices illiquidity with three defensible methods — an empirical restricted-stock estimate, Chaffe's protective-put bound, and Finnerty's average-strike model — conditions them on stage, and takes the **middle** as its central. Overriding it is a legitimate lever (you may know the company is 18 months from an IPO), but the override is recorded in the response as your assumption, not the engine's: the mark's provenance stays honest.

3.5 · The market reference — the one rule that matters

A mark is only as real as the market it is measured against, so the terminal makes the choice explicit and unmissable. **Paste your own comparables** (tickers with EV/Revenue, optional EV/EBITDA, growth, margin) and the mark is real — your data, your mark. Aim for five to fifteen names you would defend in a meeting; below five the engine warns. Choose the **synthetic reference** and every downstream number is stamped ILLUSTRATIVE, in so many words, at the response-schema level — useful for exploring mechanics, structurally incapable of masquerading as a mark. **Live comparables do not exist on the public tier** (the request refuses with a 403): redistributing fetched market data publicly is not covered by the vendor's terms, and data whose quality nobody vouched for should not price anything. Live comps run on the private tier, where someone is accountable for them.

4 · Books, controls, and reproducibility

4.1 · Building a book

Add up to 15 holdings; each keeps its own cap table, reference, and levers. The shelf shows every holding as a chip — edit or remove in place. Mixing references is allowed but never silent: if any holding rides the synthetic reference, the result carries a book-wide banner that names each illustrative holding. The banner cannot be omitted; the response format requires the names.

4.2 · Export and import — no accounts, by design

Your book exists only in your browser and, if you choose, in a JSON file on your machine (**Export book / Import book**). The file carries the form state of every holding plus the controls, under a version number the importer checks. Nothing you enter is stored server-side beyond the computation cache, request bodies are never logged, and there is no account to create because there is nothing to keep it for.

4.3 · The three stresses

toggle	exact effect
Compress multiples 30%	today's NAV: every holding's equity value scaled by 0.70 through its exact payoff curve (not the mark scaled — the curve re-evaluated); forward: all exit multiples shifted by $\ln(0.7)$ in log space
Exit window shut (~2y longer)	the copula's market factor pinned fully adverse (fewer IPOs, weaker step-ups) and every time-to-exit stretched by 1.4x
Write-off rate +10pp	the write-off probability of every holding raised by 10 percentage points

Table 4 — scenario semantics, from the engine's configuration (not paraphrased).

Stresses compose onto **all three calibration sets at once**. This is deliberate: a stress that collapsed the calibration range would be manufacturing false certainty exactly when honesty matters most. The range moves; it never closes.

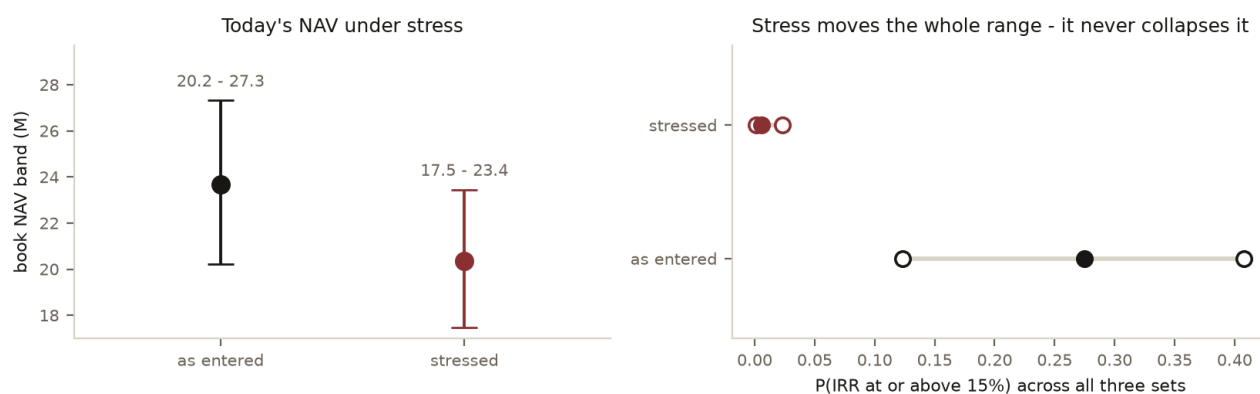


Figure 6 — the example book as entered vs. under compress-multiples + window-shut. Both the NAV band and the full calibration range shift; the disagreement between calibrations survives.

4.4 · Calibration set, seed, and paths

The **calibration set** selector (base / pessimistic / optimistic) chooses which distribution leads the headline cards and densities. It does not silence the others: the target-hit range always spans all sets. Pessimistic means harsher failure rates, weaker exit windows, later exits; optimistic the reverse; the exact shifts are printed in section 7.

The **seed** (default 20240115) makes every run bit-reproducible: same book, same controls, same seed — identical bytes back, whether computed fresh or served from cache. Change the seed to convince yourself the conclusions are not a sampling accident; if a decision flips with the seed, the honest reading is that the margin was noise. **Paths** (up to 20,000) trades runtime for Monte Carlo resolution; the default is chosen so the credible bounds are stable to well within their displayed precision.

5 · Reading the answers

5.1 · The band, seriously

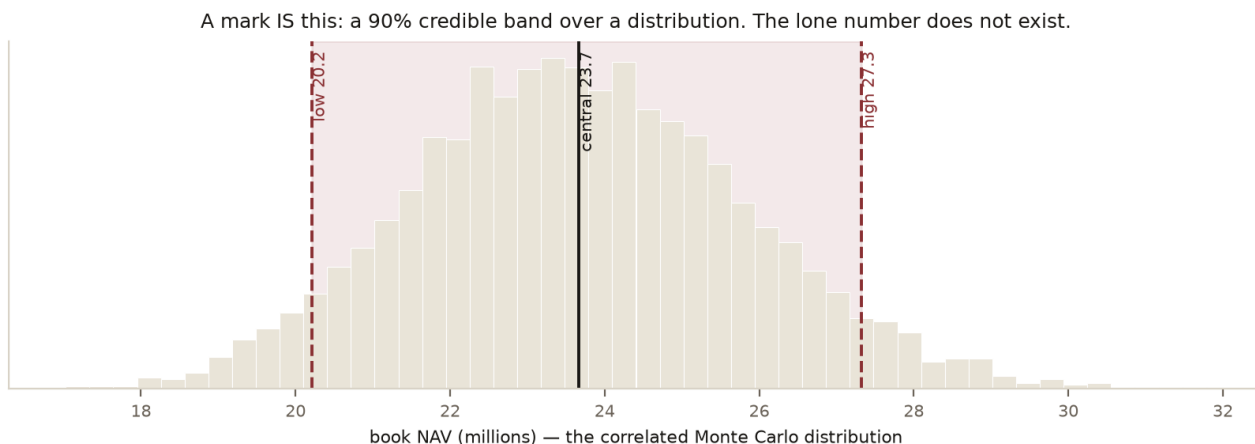


Figure 7 — the example book's NAV distribution with its 90% credible band.

A band [20.2, 27.3] at 90% means: under everything you entered and the engine's stated assumptions, 90% of the correlated Monte Carlo mass lies inside. The central is the distribution's center, **not** a promise, and not "the real number with error bars" — the band IS the answer. If you must carry one number into a spreadsheet, carry the central and write the band next to it; the terminal will never do that reduction for you.

5.2 · The four headline cards

card	how to read it
Book NAV, with its band	today's value under the correlated aggregation. Correlation honesty: the book band is generally NOT the sum of holding bands — diversification is computed, not assumed
Median TVPI	forward multiple on paid-in under the leading calibration set, with its CI and the probability of clearing the target multiple
P(IRR at or above target)	shown as a RANGE across calibration sets, whatever the selector says. IRR is deliberately secondary: annualization makes its tail unstable, so the engine reports median and interquartile range and refuses to print a mean
Effective bets	how many independent positions the book truly contains once correlation is accounted for (entropy of risk contributions). Three names can be one bet

Table 5 — the headline cards.

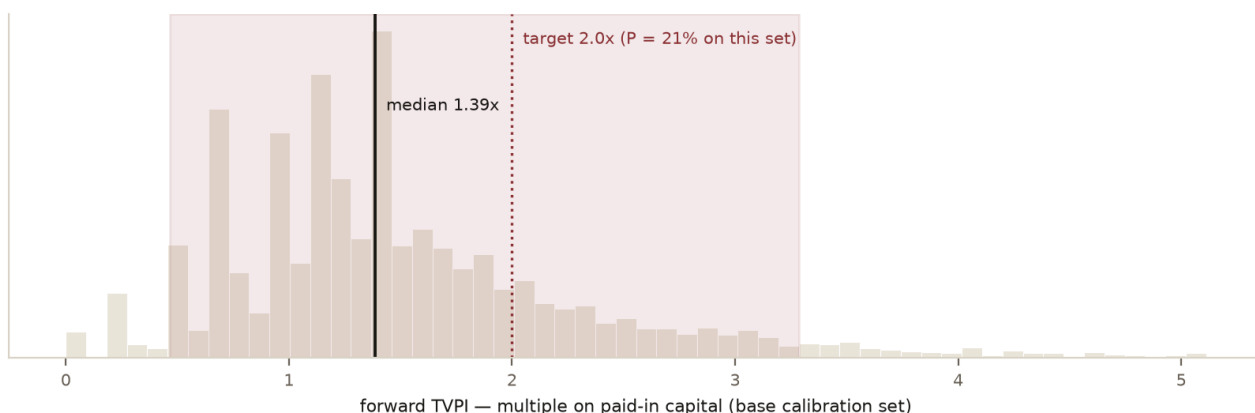


Figure 8 — the forward TVPI density (base set) with median, CI band, and target.

5.3 · The spread and the sensitivity

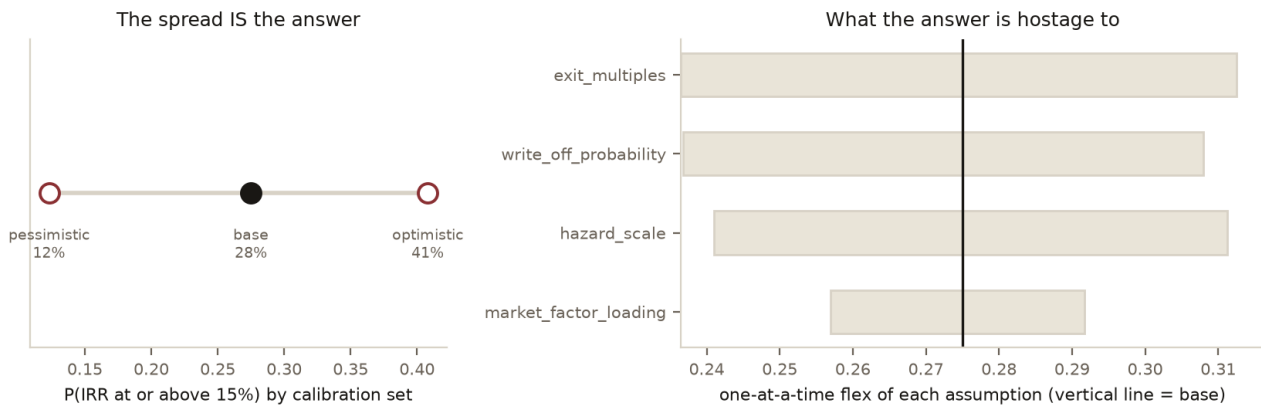


Figure 9 — left: the target-hit probability under each calibration set; the honest answer is the interval, not the dot you prefer. Right: one-at-a-time flexes of the governing assumptions, ranked — the widest bar is the assumption your conclusion is hostage to.

Use the sensitivity panel to decide what to argue about. If the bar for exit multiples spans your entire decision threshold, the productive conversation is about multiples, not about the second decimal of the median.

5.4 · The J-curve

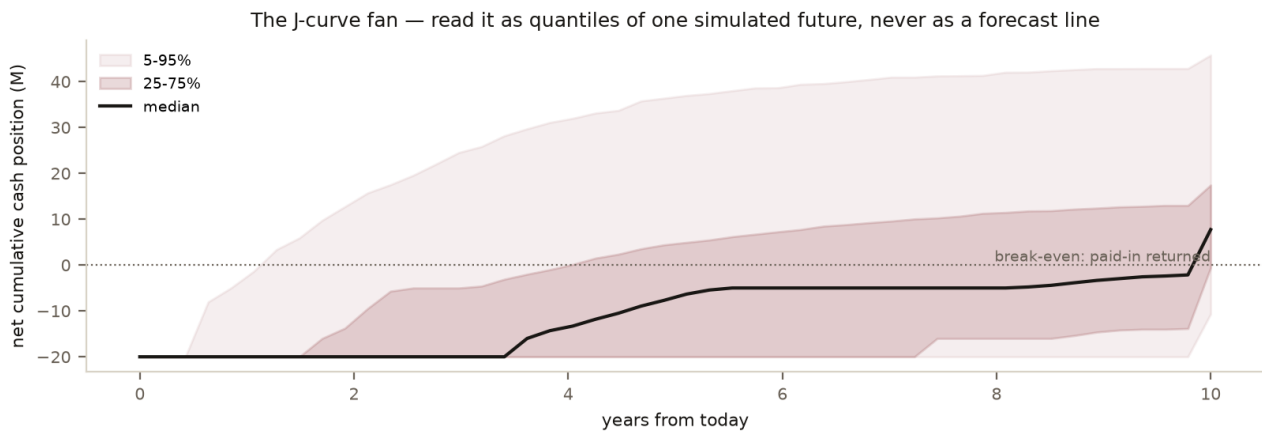


Figure 10 — the fan chart of the book's net cumulative cash position over time.

Read fans as nested quantile ribbons of one simulation, never as alternative forecasts. The deep start is the J itself — the book's paid-in capital is out from day one, and value returns only as exits arrive. The width late in the fan is the honest statement that ten-year outcomes are not knowable to a line.

5.5 · The risk view

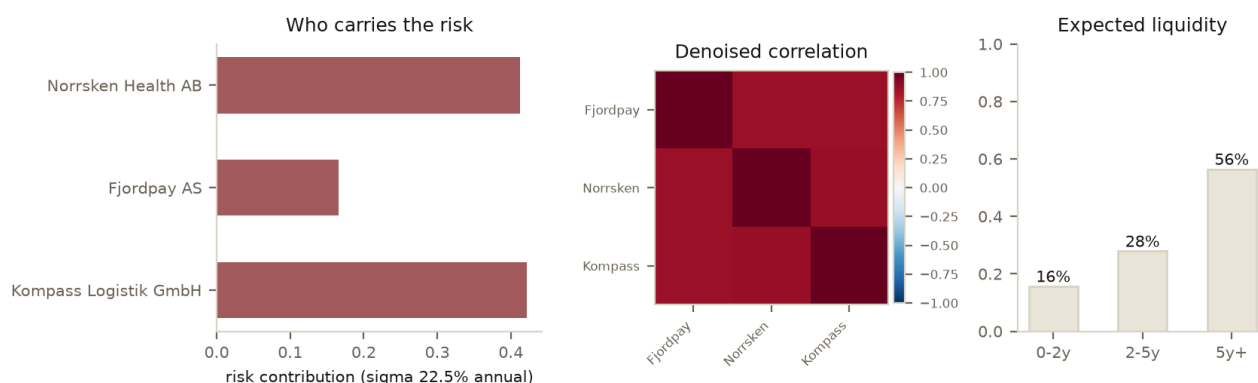


Figure 11 — risk contributions, the denoised correlation, and the liquidity ladder for the example book.

Three panels, one question each. **Who carries the risk:** Euler contributions — shares of portfolio volatility, which follow concentration and correlation, not NAV weights. **How correlated is the book:** the correlation matrix after Marchenko-Pastur denoising, which strips eigenvalues indistinguishable from sampling noise before anything is read off. **When does it turn liquid:** NAV shares expected liquid per horizon bucket, integrated from the simulated exit times.

The proxy disclosure, verbatim from the response:

"Correlation structure proxied from the reference market's sector baskets — pasted comparables carry no price history (live history is a private-tier capability)."

This matters: pasted comparables carry multiples, not return histories, so the risk view's correlations come from sector reference baskets standing in for your holdings. The note is structural — a book risk view without it cannot be emitted. Treat the risk panels as well-founded shape, not as measured covariance of your actual companies.

5.6 · Method notes and the oracle

Every response ends with the engine's method notes — the fallbacks it took, the assumptions it leaned on, in its own words — and the oracle line, e.g.:

"Every cap table in this book passed the same invariants the engine's own test suite must pass."

The oracle re-proves conservation (every distributed cent accounted), non-negativity, and monotonicity of the waterfall on your exact cap table, per request, before the mark is computed. It is paranoia as a feature: the engine checks itself in production, on your structure, not just in its test suite.

6 · When the engine says no

Refusals are part of the honesty layer. Each has one meaning:

what you see	what it means	what to do
guidance lines under the form (422)	your input is incoherent as economics — the engine explains in plain language	fix what the guidance names; nothing was computed
the access card returns (401)	no valid code on the request; the code may have been rotated	enter the current code
live comps refused (403)	public tier; live market data is private-tier only	paste comparables, or use the labeled synthetic reference
rate limit (429)	more than 30 requests in a minute from your address	wait a moment; interactive use never hits this
'Terminal offline' card	the engine is unreachable (outage or maintenance)	the demonstration still works; try again later
a run takes ~20 s	the machine was asleep (scale-to-zero) and/or the book is new to the cache	expected; repeats of the same run are instant

Table 6 — refusals and states.

Caps, and why they exist: at most 12 share classes (beyond a seed-to-pre-IPO stack), 25 comparables (a defensible peer set is 5-15), 20,000 Monte Carlo paths (double the default; bounds stop improving visibly beyond it), 15 holdings per book (bounds the cold compute at roughly a second per holding). They are protective bounds, not product tiers.

7 · What the engine computes — one honest page

The waterfall is exact, not simulated: a single pass in seniority order with conversion decided at a fixed point and participation caps resolved by exact water-filling. The payoff curve is piecewise linear with kinks located by bisection to machine tolerance; Monte Carlo paths evaluate the curve, so simulation inherits exactness.

The mark regresses your peer set's multiples on growth and margin, positions the company inside it, blends EV/Revenue and EV/EBITDA legs (your lever), walks enterprise to equity to your class through the waterfall, and applies the stage-conditional DLOM. The chain — enterprise value, equity value, pro-rata stake, waterfall stake — ships with every response.

The band comes from a correlated Monte Carlo over the drivers of that chain, at the configured 90% credibility. Correlation makes bands honest: switching it off would narrow them, which is why it cannot be switched off.

The forward simulates exit type (write-off to IPO, ordered), exit timing (age-conditional Weibull hazards), and exit size (lognormal multiples on the last-round post-money), tied together by a Gaussian one-factor copula so bad markets make everything bad at once. Three calibration sets bound the answer:

set	write-off delta	multiple shift (log)	timing scale	reading
base	0	0	1.0	the anchor calibration
pessimistic	+0.10	-0.25	1.3x slower	harsher failures, weaker windows, later exits
optimistic	-0.05	+0.20	0.8x faster	kinder failures, stronger windows, earlier exits

Table 7 — calibration sets, from configuration.

The risk model builds return proxies, denoises the correlation spectrum with Marchenko-Pastur bounds (trace preserved; cross-checked against a Ledoit-Wolf shrinkage estimate), and derives contributions, effective bets, and the liquidity ladder from the denoised structure plus simulated exit times.

8 • Boundaries: public, private, and the gate that is not this gate

The **public tier** is this terminal: pasted or loudly-labeled synthetic references, rate-limited, access-coded, no live market data. The **private tier** is a separate deployment of the same engine — not a mode, not a flag, a different server — where live comparables run behind a bearer token and a real portfolio can be wired in as a provider. There is deliberately no switch in the public codebase that could ever expose a real book.

Between the private tier and marks of record stands the **methodology gate**: independent validation of the calibrations, IPEV-alignment review, and vendor data QA. The engine's calibration honesty audit classifies every parameter as cited, anchored, or assumed — and until the assumed ones are retired or defended, output for a real book is decision support, not a mark of record. This boundary is a feature of the instrument, and it is the honest answer to "can I use this on my real portfolio today": yes to think with, not yet to report with.

9 • The owner's page

Operations live in RUNBOOK.md in the engine repository; the short version:

task	how
rotate the access code	fly secrets set PVE_ACCESS_CODE=<new> -a pve-api — takes effect on restart, seconds; the old code dies instantly
re-warm after deploy or cache clear	PVE_ACCESS_CODE=<code> python scripts/smoke_test.py https://api.myopicdelirium.com
pin the machine awake for a demo	fly scale count 1 --min 1 (undo with --min 0)
clear the cache	fly ssh console -C "rm -rf /data/api_cache" — safe anytime; exact cache rebuilds on demand
logs	fly logs — method, path, status, milliseconds; request bodies are never logged

Table 8 — the operational surface. The access code itself is stored only as a Fly secret; it appears nowhere in code, configuration files, or this manual.

Cost posture: one small machine in Stockholm, scale-to-zero, a 1 GB volume for the cache — with the gate keeping drive-by traffic off the compute path, on the order of one to two dollars a month.

Appendix A • The book file format

An exported book is a JSON object: **version** (currently 1 — the importer refuses other versions), **book** (an array of holdings exactly as the form holds them: company, classes, stake, lastRound, refMode, comps, and the two lever overrides, all as entered), and **controls** (the three stress toggles, the calibration set, the seed, and paths). It is plain text: version it, diff it, mail it. Import replaces the current shelf and restores the controls; press run and, at the same seed, you get the same bytes back.

Appendix B • Glossary

term	meaning
band / credible interval	low-central-high containing the stated share (90%) of Monte Carlo mass under the given assumptions
calibration set	a named, internally consistent set of exit-model parameters (base / pessimistic / optimistic)

term	meaning
DLOM	discount for lack of marketability — the price of not being able to sell
effective bets	entropy-based count of independent risk positions in the book
gap (driver)	engine central minus your self-mark, decomposed into comps and DLOM parts; the driver names the larger
illustrative	computed against the synthetic reference; mechanics only, never a mark — always labeled
oracle	the per-request re-proof of waterfall invariants on your exact cap table
preference threshold	the exit equity value at which the stack's liquidation preferences are exactly covered
seed	the integer fixing all randomness; same seed + same inputs = identical output, bit for bit
TVPI	total value to paid-in — the multiple on invested capital; the engine's preferred forward headline
waterfall	the exact allocation of exit proceeds down the cap table by seniority, preference, participation, conversion

Built from the live engine · every figure and quoted number regenerated at build time · seed 20240115 · the honesty layer is the product.